Machine Learning Models for County-Level Analysis of Voter Turnout and Partisan Lean in the 2024 U.S. Presidential Election

Isak Dai, Courtney Green, Jonah Lichtenthal
Georgetown University Data Science & Analytics

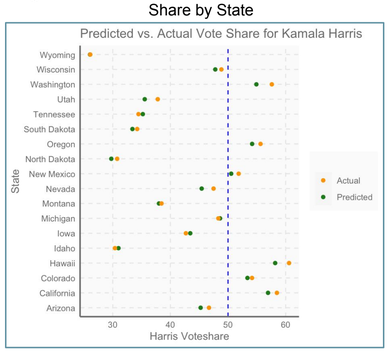
Introduction

This project explores predictive modeling of U.S. county-level outcomes in the 2024 presidential election, using demographic, socioeconomic, and historical election data. The goal is to simulate real-time election night forecasting by training models on early-reporting states and testing on those that report later. This setup mirrors the decision-making environment faced by media outlets and analysts during live coverage. The dataset integrates county-level data on race, education, income, poverty, unemployment, and population, along with 2020 and 2024 vote shares. The models aim to (1) predict Democratic vote share, (2) classify Democratic wins, (3) estimate total turnout, and (4) identify key predictors of partisan lean.

Data Preprocessing

Demographic, socioeconomic, and election data were compiled from multiple sources; race data came from the 2023 American Community Survey (ACS). Poverty, education, unemployment, income, and rural-urban status were obtained from the USDA Economic Research Service. County-level election results for 2020 and 2024 were scraped from public sources, including *Fox News, Politico*, and *The New York Times*. All datasets were cleaned, standardized, and merged using FIPS codes. States were manually split into training and test sets to reflect a staged release of results, simulating real-time prediction conditions.

Figure 1. Predicted vs. Actual 2024 Democratic Vote Share by State



Linear Regression

Four linear regression variants were used to predict Kamala Harris' 2024 county-level vote share and total turnout: (1) standard linear regression, (2) linear regression with 5-fold cross-validation, (3) best subset selection via BIC, and (4) LASSO regression with cross-validated lambda. The first three offer interpretable coefficients; LASSO adds regularization, shrinking less relevant variables, though at risk of underfitting.

Table 1. Performance Metrics for Regression Models Predicting Democratic Vote Share

Regression Model – Partisan Lean	MAE	RMSE
Linear Regression	.0113	.0145
Linear Regression with CV	.0110	.0143
Best Subset	.0111	.0143
LASSO	.0110	.0141

Table 2. Performance Metrics for Regression Models Predicting Voter Turnout

Regression Model – Turnout	MAE	RMSE
Linear Regression	3,447	12,095
Linear Regression with CV	3,668	11,821
Best Subset	11,271	25,199
LASSO	12,212	3,599

Artificial Neural Network (ANN)

An ANN was trained for three tasks: (1) predicting Democratic vote share (regression), (2) classifying Democratic county-level wins, and (3) predicting total votes cast. For classification, a binary target was created based on whether *per_dem_24* exceeded 50%. The network used two hidden layers with ReLU, dropout, batch normalization, and L2 regularization. Early stopping and the Adam optimizer were used. Evaluation included accuracy, precision, recall, F1 score, and confusion matrix.

Figure 8. Confusion Matrix of ANN Predictions for 2024 County Winners

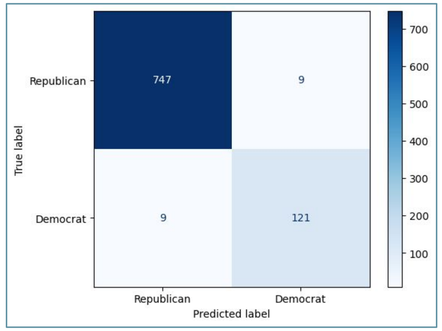


Figure 3. KNN Validation Curve

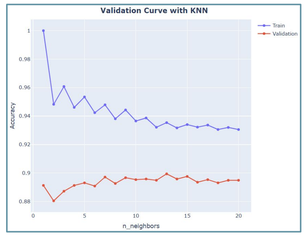


Table 3. Performance Metrics for Classification Models Predicting Democratic Leaning Counties

Classification Models – Partisan Lean	Accuracy	Precision	Recall	F1 Score	MAE
ANN	0.981	.970	.970	.969	.017
KNN	.867	.928	.844	.884	.133
Logistic Regression	.970	.972	.977	.975	.030
Random Forest	.974	.970	.987	.979	.026
SVM	.951	.979	.940	.959	.049

Binary Classification

A binary classification task was conducted to predict whether a county leaned Democratic in the 2024 election. Counties were labeled 1 if their Democratic vote share (*per_dem_24)* exceeded the median of the training set. Features were standardized using z-scores, and predictors were selected using best subset selection. Four classification algorithms were applied:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

Models were evaluated using accuracy, precision, recall, F1 score, and MAE. Validation curves were generated using 5-fold cross-validation.

Figure 4. Logistic Regression Validation Curve

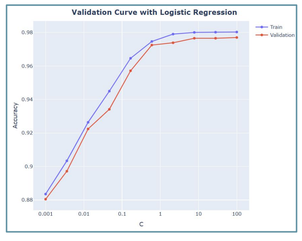
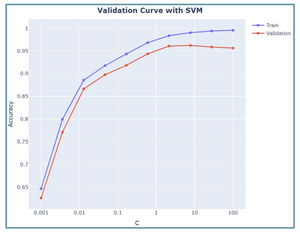


Figure 5. SVM Validation Curve



Results

For the Democratic-Leaning classification task **(Table 3),** all models performed well, but Random Forest and Logistic Regression offered the best balance of recall and interpretability. Random Forest **(Figure 6)** achieved the highest recall (0.987), correctly identifying the most Democratic-leaning counties. SVM delivered the highest precision (0.979), minimizing false positives. Logistic Regression **(Figure 4)** remained easy to interpret while maintaining strong performance. The Artificial Neural Network (ANN) achieved the highest overall accuracy (0.981) and lowest MAE (0.017), though at the cost of transparency.

The feature importance plot from the Random Forest model **(Figure 7)** shows Biden's 2020 vote share as the dominant driver of predictions. Other impactful features included racial composition (e.g., percent Black), population, and 2024 total turnout. Education and poverty contributed little, suggesting that prior voting patterns and demographics were the primary signals guiding the model.

For predicting Democratic vote share as a continuous outcome **(Table 1),** LASSO and cross-validated linear regression yielded the lowest MAE (0.0110). All models had low RMSEs, indicating strong fit. Linear models were particularly effective due to their interpretability and competitive performance.

LASSO interpretation confirmed Biden's 2020 vote share as the strongest predictor of Harris' 2024 support, showing a near 1:1 relationship. Counties with higher net migration and unemployment were modestly associated with lower Democratic support, reflecting broader demographic and economic shifts.

For turnout prediction **(Table 2),** 2020 turnout was the strongest predictor. However, total population had a negative correlation with 2024 turnout—suggesting that, after adjusting for size, larger counties turned out at lower rates. This trend aligns with the observed negative relationship between urbanicity and turnout.

A comparison of predicted vs. actual Democratic vote share by state **(Figure 1)** further illustrates the model's accuracy. Predicted values closely match actual outcomes, with swing states like Wisconsin and Michigan called correctly,, demonstrating the linear regression model's reliability in real-time estimation scenarios.

Conclusion

Political organizations and the media alike have an incentive in predicting the winner of an election based on partial returns as early and accurately as possible. As a result, it is pertinent to study which model best predicts each county's turnout and partisan lean based on partial results that are reported earlier in the night. In other words, can data from states that report returns early in the night (e.g., Kentucky, Indiana, and Florida) be used to predict results in swing states that report late in the night? How accurate are these predictions?

Linear regression models performed best in this scenario. The best-performing linear regression models (LASSO for partisan lean and cross-validated linear regression for turnout) had lower errors, on average, than a more complex artificial neural network. The end result was a combined turnout and partisan lean model that correctly predicted the winner of each of the states in our test set.

In classification tasks, an artificial neural network had the highest accuracy (98.1%) than other tested methods like k-nearest neighbors, logistic regression, or random forest. This suggests a more complex relationship between the predictor variables and whether or not a county had a higher-than-median Democratic vote share or turnout.

Figure 6. Random Forest Validation Curve

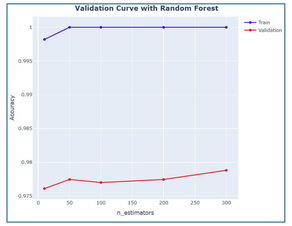
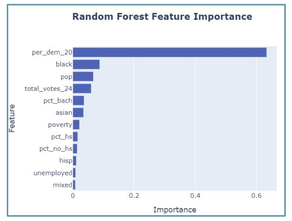


Figure 7. Feature Importance from Random Forest Model



Discussion

Although model performance was strong across tasks and models, especially in predicting the Democratic vote share, the study had key limitations. The 2024 election data were obtained from scraped results, including those from *Fox News* for 2024 and *Fox News, Politico, and The New York Times* for 2020. These are not official datasets and may include missing or inconsistent county-level data. This affects data quality and limits reliability across states. The classification tasks employed median splits to define binary targets, which simplified the problem but also removed detail and may have obscured patterns in more competitive or marginal counties.

Future versions of this project could incorporate updated and verified 2024 election results once official data becomes available. Expanding the dataset to include more election cycles or modeling third-party vote shares may also uncover longer-term patterns. Moreover, experimenting with ensemble models or deep learning approaches could help improve performance in more competitive or noisy counties.

References

McGovern, T. (2025). *United States General Election Presidential Results by District, Ward, or County from 2008 to 2024.* https://github.com/tonmcg/US_County_Level_Election_Results_08-24/tree/master?tab=MIT-1-ov-file

U.S. Department of Agriculture, Economic Research Service. (2025). *County-level data sets.* https://www.ers.usda.gov/data-products/county-level-data-sets